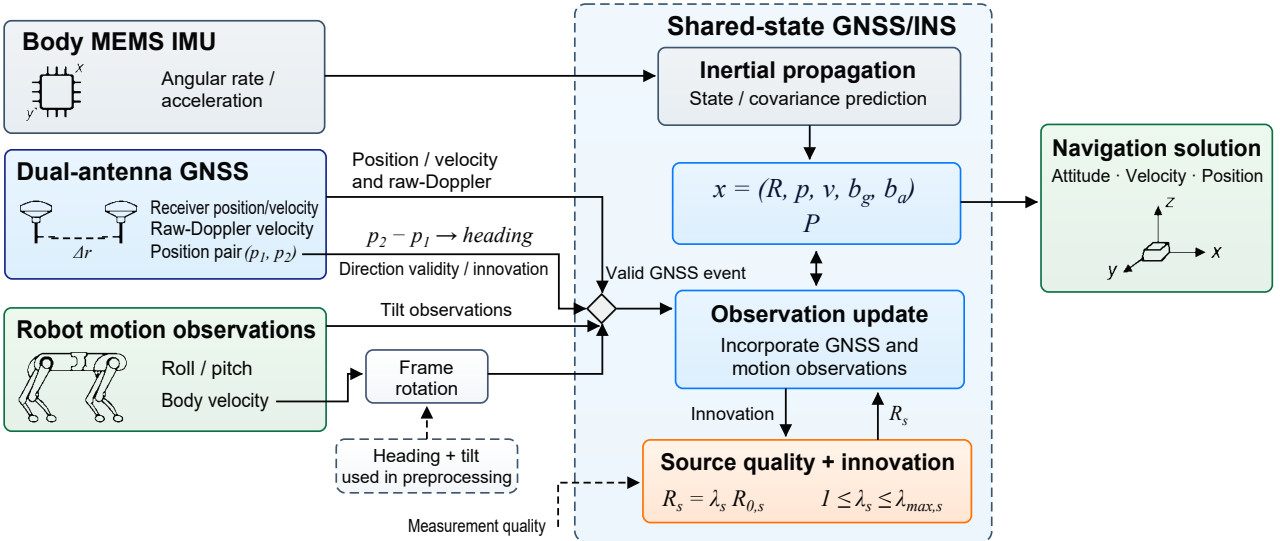

Body MEMS IMU
x
y
Angular rate / acceleration
Dual-antenna GNSS
Δr
Receiver position/velocity
Raw-Doppler velocity
Position pair (p1, p2)
Robot motion observations
Roll / pitch
Body velocity
Position / velocity and raw-Doppler
p2 − p1 → heading
Direction validity / innovation
Tilt observations
Frame rotation
Heading + tilt used in preprocessing
Measurement quality
Shared-state GNSS/INS
Inertial propagation
State / covariance prediction
x = (R, p, v, bg, ba)
P
Valid GNSS event
Observation update
Incorporate GNSS and motion observations
Innovation
Rs
Source quality + innovation
Rs = λs R0,s
1 ≤ λs ≤ λmax,s
Navigation solution
Attitude · Velocity · Position
z
x
y